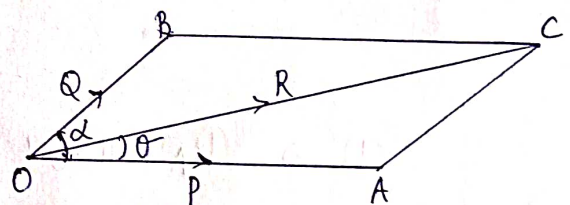
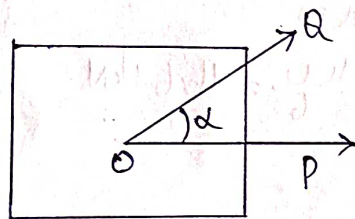
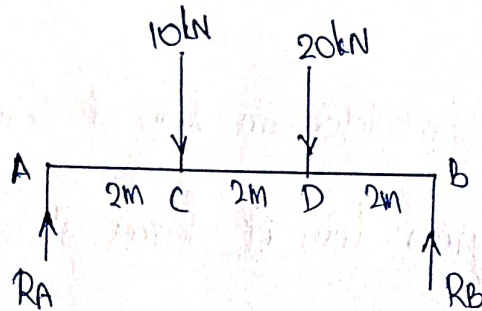
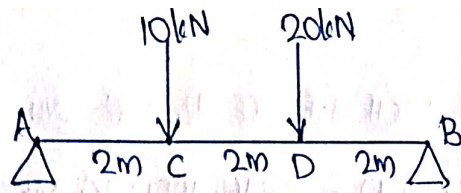


1. State and explain parallelogram law of forces.

Parallelogram law of forces states that if two forces acting simultaneously on a body at a point are represented in magnitude and direction by the adjacent sides of a parallelogram, then the resultant is represented in magnitude and direction by the diagonal at the point of intersection of the two sides representing the forces.



2. A simply supported beam AB of span 6m is carrying point loads 10kN & 20kN at 2m & 4m respectively from support A. Calculate reactions at supports A and B.



$$\sum F_x = 0$$

$$\sum F_y = 0 \Rightarrow R_A + R_B - 10 - 20 = 0$$

$$R_A + R_B = 30 \quad \text{--- (1)}$$

$$\sum M_A = 0 \Rightarrow -(10 \times 2) - (20 \times 4) + (R_B \times 6) = 0$$

$$= -20 - 80 + 6R_B = 0$$

$$6R_B = 100$$

$$R_B = \frac{100}{6} = \underline{\underline{16.67 \text{ kN}}}$$

$$(1) \Rightarrow R_A + 16.67 = 30$$

$$R_A = 30 - 16.67$$

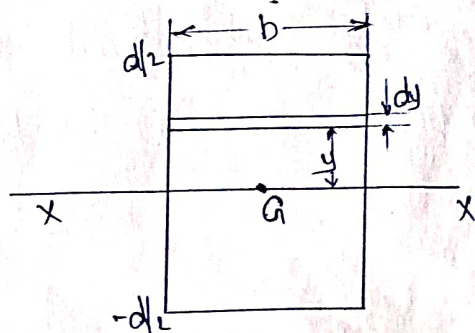
$$R_A = \underline{\underline{13.33 \text{ kN}}}$$

(2)

3. Explain Coulomb's law of friction.

- The frictional force always acts in a direction opposite to that in which the body moves or tends to move
- Till the limiting value is reached, the magnitude of frictional force is equal to applied force which tends to move the body
- The magnitude of frictional force bears a constant ratio to the normal reaction between the two surfaces of contact and this ratio is called co-efficient of friction.
- The force of friction depends upon the roughness of surface in contact.

4. Calculate the moment of inertia of rectangular cross section of breadth b , and depth d about the centroidal horizontal axis.



Moment of inertia with respect to centroidal axis,

$$I_{Cxx} = \int dA \cdot y^2$$

$$= \int_{-d/2}^{d/2} (b \cdot dy) \cdot y^2$$

$$= b \cdot \int_{-d/2}^{d/2} y^2 dy$$

$$= b \left[\frac{y^3}{3} \right]_{-d/2}^{d/2}$$

$$= \frac{b}{3} \left[\frac{d^3}{8} - \frac{-d^3}{8} \right]$$

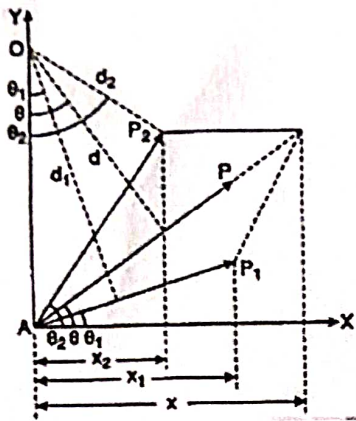
$$= \frac{b}{3} \left[\frac{d^3}{8} + \frac{d^3}{8} \right]$$

$$= \frac{b}{3} \times \frac{2d^3}{8}$$

$$\underline{\underline{I_{ax1} = \frac{bd^3}{12}}}$$

5. State and explain Varignon's theorem of moments.

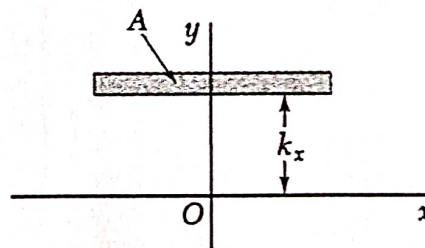
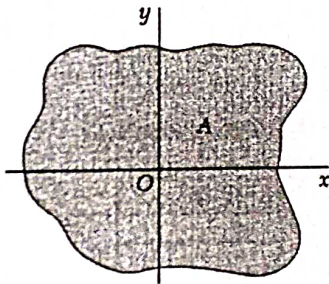
It states that the moment of the resultant about any point is equal to the algebraic sum of moments of all forces of the system about the same point.



$$P \times d = P_1 \times d_1 + P_2 \times d_2$$

6. Explain radius of gyration of an area.

Consider area A with moment of inertia I_x , Imagine that the area is concentrated in a thin strip parallel to the x axis with equivalent I_x



$$I_x = k_x^2 A \quad k_x = \sqrt{\frac{I_x}{A}}$$

k_x = radius of gyration with respect to the x axis

Radius of Gyration of a body (or a given lamina) about an axis is the distance such that its square multiplied by the area gives M.I of the area about the given axis.

7. Explain D'Alembert's principle.

The system of forces acting on a moving body is in dynamic equilibrium with the inertia force of the body. It is also known as equation of dynamic equilibrium.

$$F - ma = 0$$

8. Explain work energy principle in dynamics

Work energy principle states that the total work done by the sum of all the forces acting on a particle is equal to the change in kinetic energy of that particle.

$$F = ma$$

$$= m \frac{dv}{dt}$$

$$= m \cdot \frac{dv}{dt} \cdot \frac{ds}{ds}$$

$$= m \cdot \frac{ds}{dt} \cdot \frac{dv}{ds}$$

$$F \cdot ds = m \cdot v \cdot dv$$

$$\int_0^s F \, ds = \int_u^v m v \, dv$$

$$F \cdot [s]_0^s = m \left[\frac{v^2}{2} \right]_u^v$$

$$F \cdot s = \frac{mv^2}{2} - \frac{mu^2}{2}$$

Work done = Change in kinetic energy

9. Distinguish damped and undamped free vibrations.

Damped

Undamped

→ The oscillations whose amplitude decreases gradually with time

→ The oscillations whose amplitude remains constant with time

→ Does not continue for a longer time

→ There are no power losses

→ Will die out eventually

→ Will oscillate indefinitely.

10. A particle executing SHM passes through two points P and Q that are 20 cm apart, with the same velocity in a time interval of 4 seconds. If the particle returns to Q after another 2 seconds, find the period and amplitude of motion.

$$T = 12 \text{ s}$$

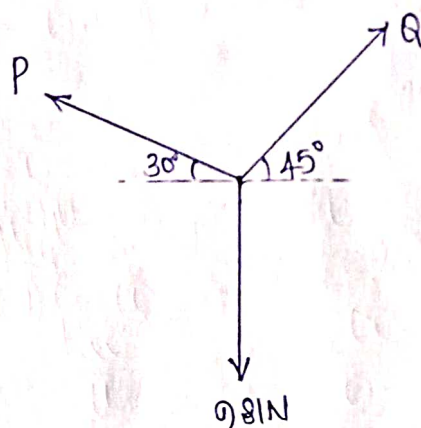
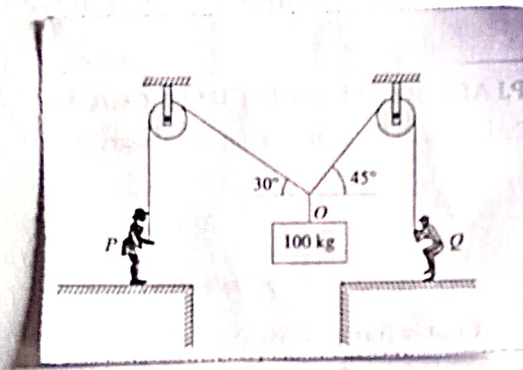
$$\omega = \frac{2\pi}{12} = \frac{\pi}{6} \text{ rad/s}$$

~~$$x = r \sin(\omega t)$$~~

$$x = r \sin(\omega t)$$

11.

- a. Two persons lift a mass of 100 kg by cables passing over two pulleys as shown in figure. Determine the forces P and Q that must be applied by the two persons if the body is in equilibrium at the position shown.



By applying Lami's theorem,

$$\frac{981}{\sin 105} = \frac{P}{\sin 135} = \frac{Q}{\sin 120}$$

$$P = \frac{981 \times \sin 135}{\sin 105} = \underline{\underline{718.14 \text{ N}}}$$

$$Q = \frac{981 \times \sin 120}{\sin 105} = \underline{\underline{879.51 \text{ N}}}$$

- b. The greatest and least resultants of two forces F_1 and F_2 are 17 N and 3 N respectively. Determine the angle between them when their resultant is 14 N .

$$F_1 + F_2 = 17 \text{ N}$$

$$F_1 - F_2 = 3 \text{ N}$$

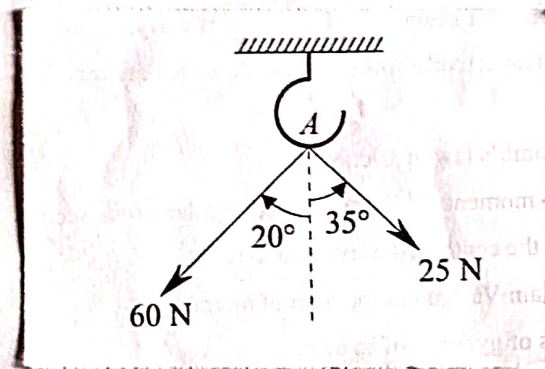
$$2F_1 = 20 \text{ N}$$

$$F_1 = \underline{\underline{10 \text{ N}}}$$

$$F_2 = \underline{\underline{7 \text{ N}}}$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

12. a. Two forces are applied at the point A of a hook support as shown in figure. Determine the magnitude and direction of resultant force.



$$P = 25 \text{ N}$$

$$Q = 60 \text{ N}$$

$$\alpha = 55^\circ$$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

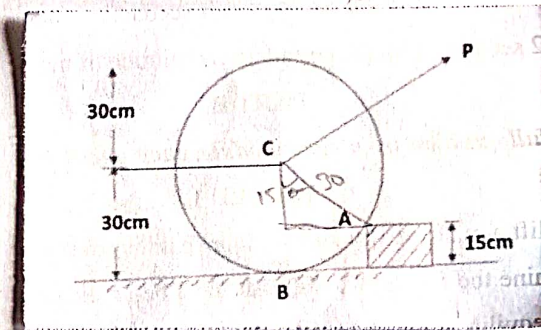
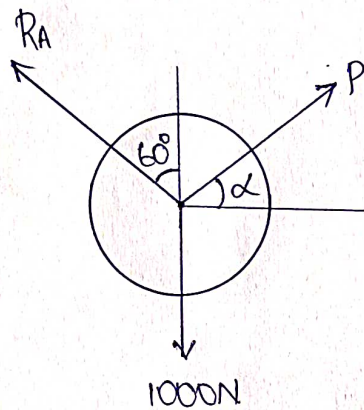
$$= \sqrt{25^2 + 60^2 + 2 \times 25 \times 60 \times \cos 55}$$

$$= \underline{\underline{77.11 \text{ N}}}$$

$$\tan \theta = \frac{Q \sin \alpha}{P + Q \cos \alpha} = \frac{60 \times \sin 55}{25 + 60 \cos 55} = 0.82$$

$$\theta = 39.59^\circ \text{ with the force of } 25 \text{ N}$$

- b. A uniform wheel 60cm diameter weighing 1000N rests against a rectangular obstacle 15cm height as shown in figure. Find the least force required, which when acting through the centre of wheel will just turn the wheel over the corner A of the block. Also find the reaction of block.



From the given figure, the angle between the R_A and vertical is 60° . Then angle between P and horizontal took it as α .

By applying Lami's theorem,

$$\frac{1000}{\sin(60+90-\alpha)} = \frac{P}{\sin 120} = \frac{R_A}{\sin(90+\alpha)}$$

$$P = \frac{1000 \sin 120}{\sin(90-\alpha)}$$

$$P_{\min} = \frac{1000 \sin 120}{\sin 90}$$

For getting minimum value of P , denominator should be maximum, i.e. $\sin 90 = 1$

$$\therefore P_{\min} = \frac{1000 \times \sin 120}{\sin(150 - \alpha)} = \frac{1000 \times \sin 120}{\sin(150 - 60)}$$

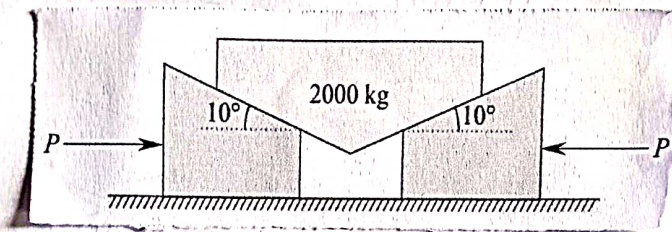
$$P_{\min} = \frac{1000 \times \sin 120}{\sin 90} = \underline{\underline{866.02 \text{ N}}}$$

$$R_A = \frac{1000 \times \sin(90 + \alpha)}{\sin(60 + 90 - \alpha)}$$

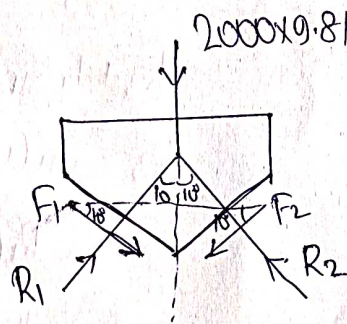
$$= \frac{1000 \times \sin(90 + 60)}{\sin(60 + 90 - 60)} = \frac{1000 \times \sin 150}{\sin 90}$$

$$R_A = \underline{\underline{500 \text{ N}}}$$

13. A 2000kg block must be raised using two similar wedges as shown in figure. What horizontal force P must be applied on both the wedges to raise the block? Take coefficient of friction at all contact surfaces is 0.2.



FBD of 2000kg block,



$$\sum F_x = 0$$

$$R_1 \sin 10^\circ + F_1 \cos 10^\circ - R_2 \sin 10^\circ - F_2 \cos 10^\circ = 0$$

$$R_1 \sin 10^\circ + 0.2 R_1 \cos 10^\circ = R_2 \sin 10^\circ + 0.2 R_2 \cos 10^\circ$$

$$R_1 (\sin 10^\circ + 0.2 \cos 10^\circ) = R_2 (\sin 10^\circ + 0.2 \cos 10^\circ)$$

$$R_1 = R_2$$

$$\underline{\Sigma F_y = 0}$$

$$R_1 \cos 10^\circ + R_2 \cos 10^\circ - F_1 \sin 10^\circ - F_2 \sin 10^\circ - 19620 = 0$$

$$R_1 \cos 10^\circ + R_1 \cos 10^\circ - 0.2 R_1 \sin 10^\circ - 0.2 R_1 \sin 10^\circ = 19620$$

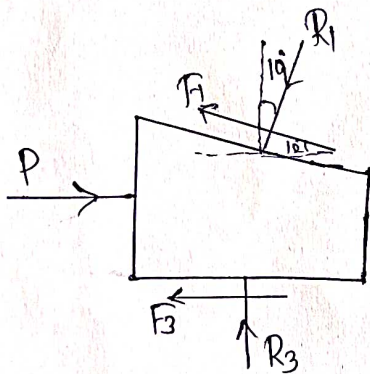
$$2R_1 \cos 10^\circ - 0.4 R_1 \sin 10^\circ = 19620$$

$$R_1 (1.969 - 0.069) = 19620$$

$$R_1 = \frac{19620}{1.9}$$

$$= \underline{\underline{10326.31 \text{ N}}}$$

FBD of wedge



$$\underline{\Sigma F_y = 0}$$

$$R_3 + F_1 \sin 10^\circ - R_1 \cos 10^\circ = 0$$

$$R_3 = R_1 \cos 10^\circ - 0.2 R_1 \sin 10^\circ$$

$$= R_1 (\cos 10^\circ - 0.2 \sin 10^\circ)$$

$$= 10326.31 \times 0.95$$

$$= \underline{\underline{9809.99 \text{ N}}}$$

$$\underline{\Sigma F_x = 0}$$

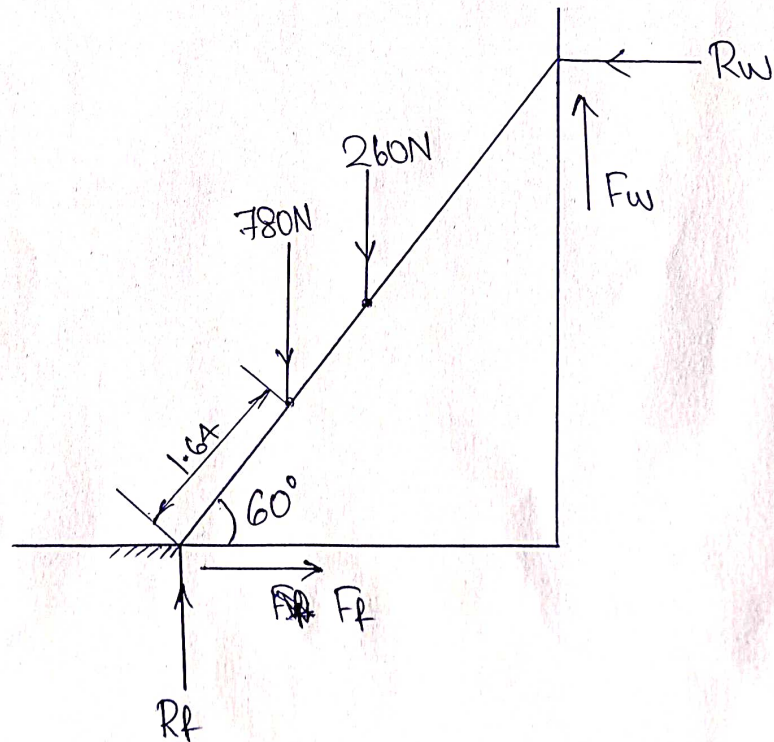
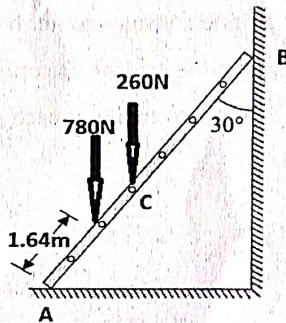
$$P - F_3 - F_1 \cos 10^\circ - R_1 \sin 10^\circ = 0$$

$$P = R_1 \sin 10^\circ + 0.2 R_1 \cos 10^\circ + 0.2 R_3$$

$$= 1793.14 + 2033.88 + 1961.99$$

$$= 5789.01 \text{ N} = \underline{\underline{5.78 \text{ kN}}}$$

14. A ladder 5m long and weighing 260N, is placed against a vertical wall at an inclination of 30° with wall. A man weighing 780N climbs the ladder. When he is at a distance of 1.64m along ladder from lower end, ladder slips. What is the reaction at wall, floor and coefficient of friction, assuming the same for all contact surface.



Apply conditions of equilibrium,

$$\underline{\sum F_x = 0}$$

$$F_f - R_w = 0$$

$$F_f = R_w$$

$$\mu R_f = R_w \quad \text{--- (1)}$$

$$\underline{\sum F_y = 0}$$

$$R_f + F_w - 780 - 260 = 0$$

$$R_f + \mu R_w = 1040$$

$$\frac{R_w}{\mu} + \mu R_w = 1040 \Rightarrow R_w + \mu^2 R_w = 1040\mu$$

$$R_w(1 + \mu^2) = 1040\mu \Rightarrow R_w = \frac{1040\mu}{1 + \mu^2} \quad \text{--- (2)}$$

$$\underline{\sum M_f = 0}$$

$$-(780 \times 1.64 \cos 60) - (260 \times 2.5 \cos 60) - (F_w \times 5 \cos 60) + (R_w \times 5 \sin 60) = 0$$

$$= -(780 \times 0.82) - (260 \times 1.25) + (\mu R_w \times 2.5) + (R_w \times 4.33) = 0$$

$$= R_w(2.5\mu + 4.33) = 639.6 + 325$$

$$= R_w(2.5\mu + 4.33) = 964.6$$

$$= \frac{1040\mu}{(1+\mu^2)} (2.5\mu + 4.33) = 964.6$$

$$= \frac{2600\mu^2}{1+\mu^2} + \frac{4503.2\mu}{1+\mu^2} = 964.6$$

$$= 2600\mu^2 + 4503.2\mu = 964.6(1+\mu^2)$$

$$= 2600\mu^2 + 4503.2\mu = 964.6 + 964.6\mu^2$$

$$= 1635.4\mu^2 + 4503.2\mu - 964.6 = 0$$

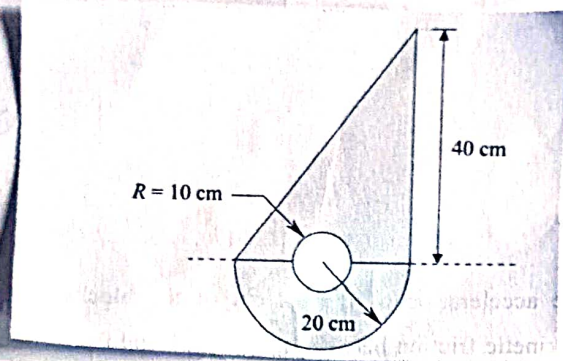
By solving this equation,

$$\mu = 0.199 \approx \underline{\underline{0.2}}$$

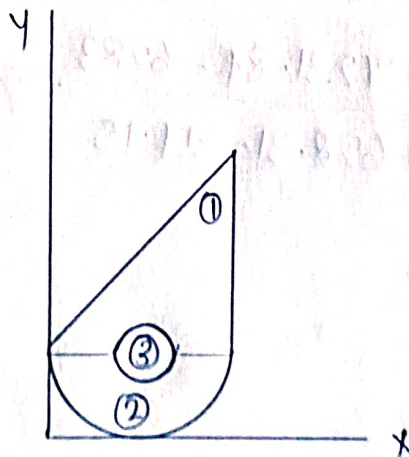
$$R_w = \frac{1040\mu}{1+\mu^2} = \frac{1040 \times 0.2}{1+(0.2)^2} = \underline{\underline{200N}}$$

$$R_f = \frac{R_w}{\mu} = \frac{200}{0.2} = \underline{\underline{1000N}}$$

15. Find the moment of inertia of shaded area about the centroidal axes

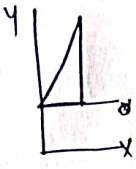
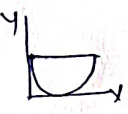


To find the centroid of given area,



$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 - A_3 x_3}{A_1 + A_2 - A_3} = \frac{\sum A_i x_i}{\sum A_i}$$

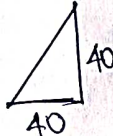
$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 - A_3 y_3}{A_1 + A_2 - A_3} = \frac{\sum A_i y_i}{\sum A_i}$$

SL No.	Fig	Area(A _i)	x _i	y _i	A _i x _i	A _i y _i
1.		$\frac{1}{2}bh$ $= \frac{1}{2} \times 40 \times 40$ $= 800$	$\frac{2b}{3}$ $= \frac{2 \times 40}{3}$ $= 26.66$	$\frac{2h}{3}$ $= \frac{2 \times 40}{3}$ $= 33.33$	21328	26664
2.		$\frac{\pi r^2}{2}$ $= \frac{\pi \times 20^2}{2}$ $= 628.31$	$r = 20$	$r - \frac{4r}{3\pi}$ $= 20 - \frac{4 \times 20}{3\pi}$ $= 11.51$	12566.2	7231.84
3.		πr^2 $= \pi \times 10^2$ $= 314.15$	$\frac{10+r}{2}$ $= \frac{10+10}{2}$ $= 20$	$\frac{10+r}{2}$ $= \frac{10+10}{2}$ $= 20$	6283	6283

$$\bar{x} = \frac{21328 + 12566.2 - 6283}{800 + 628.31 - 314.15} = \underline{\underline{24.78 \text{ cm}}}$$

$$\bar{y} = \frac{26664 + 7231.84 - 6283}{800 + 628.31 - 314.15} = \underline{\underline{24.78 \text{ cm}}}$$

For finding moment of inertia,

SL No.	Fig	Area	I_{xx}	I_{yy}	$y_1 - \bar{y}$	$x_1 - \bar{x}$	$A(dy)^2$	$A(dx)^2$
1.		800	$\frac{bh^3}{36}$ $= \frac{40 \times 40^3}{36}$ $= 71111.11$	$\frac{hb^3}{36}$ $= \frac{40 \times 40^3}{36}$ $= 71111.11$	$y_1 - \bar{y}$ $= 33.3 - 24.78$ $= 8.55$	$x_1 - \bar{x}$ $= 26.66 - 24.78$ $= 1.88$	58482	2827.5
2.		628.31	$\frac{\pi r^4}{8}$ $= \frac{\pi \times 20^4}{8}$ $= 17600$	$\frac{\pi D^4}{128}$ $= \frac{\pi \times 40^4}{128}$ $= 62831.8$	$y_2 - \bar{y}$ $= 11.51 - 24.78$ $= -13.27$	$x_2 - \bar{x}$ $= 20 - 24.78$ $= -4.78$	110640.9	14355.8
3.		314.15	$\frac{\pi D^4}{64}$ $= \frac{\pi \times 20^4}{64}$ $= 7853.98$	$\frac{\pi D^4}{64}$ $= 7853.98$	$y_3 - \bar{y}$ $= 20 - 24.78$ $= -4.78$	$x_3 - \bar{x}$ $= 20 - 24.78$ $= -4.78$	7177.82	7177.82

$$\sum I_{xx} = 71111.11 + 17600 - 7853.98 = 80857.13$$

$$\sum I_{yy} = 71111.11 + 62831.8 - 7853.98 = 126088.93$$

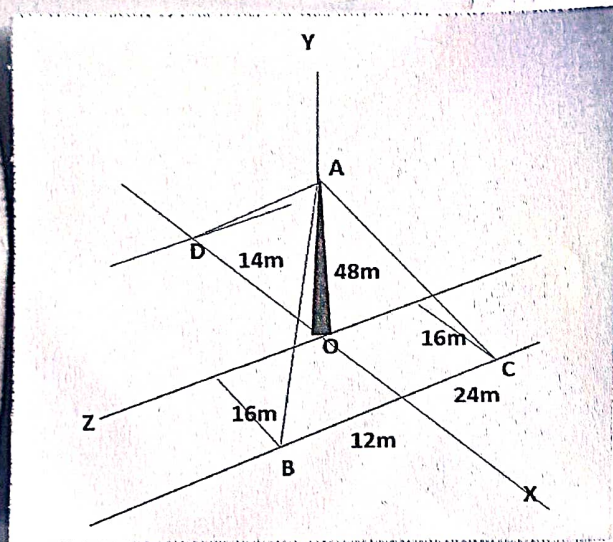
$$\sum A(dy)^2 = 58482 + 110640.9 - 7177.82 = 161945.08$$

$$\sum A(dx)^2 = 2827.52 + 14355.8 - 7177.82 = 10005.5$$

$$I_{xx} = \sum I_{xx} + \sum A(dy)^2 = \underline{\underline{242802.21 \text{ cm}^4}}$$

$$I_{yy} = \sum I_{yy} + \sum A(dx)^2 = \underline{\underline{136094.43 \text{ cm}^4}}$$

16. A post is held in vertical position by three cables AB, AC and AD as shown in figure. If the tension in the cable AB is 40N, calculate the required tension in AC and AD so that the resultant of the three forces applied at A is vertical.



From the given figure,

$$A(0, 48, 0)$$

$$B(16, 0, 12)$$

$$C(16, 0, -24)$$

$$D(-14, 0, 0)$$

Now the vector form of forces,

$$\vec{F}_{AB} = |F_{AB}| \cdot \lambda_{AB}$$

$$= 40 \left(\frac{16\hat{i} - 48\hat{j} + 12\hat{k}}{\sqrt{16^2 + 48^2 + 12^2}} \right)$$

$$= 12.31\hat{i} - 36.92\hat{j} + 9.23\hat{k}$$

$$\vec{F}_{AC} = |F_{AC}| \cdot \lambda_{AC}$$

$$= F_{AC} \left(\frac{16\hat{i} - 48\hat{j} - 24\hat{k}}{\sqrt{16^2 + 48^2 + 24^2}} \right)$$

$$= 0.29 F_{AC} \hat{i} - 0.86 F_{AC} \hat{j} - 0.43 F_{AC} \hat{k}$$

$$\vec{F}_{AD} = |F_{AD}| \cdot \lambda_{AD}$$

$$= F_{AD} \left(\frac{-14\hat{i} - 48\hat{j} - 0\hat{k}}{\sqrt{14^2 + 48^2}} \right)$$

$$= -0.28 F_{AD} \hat{i} - 0.96 F_{AD} \hat{j} + 0\hat{k}$$

Resultant,

$$\vec{R} = \vec{F}_{AB} + \vec{F}_{AC} + \vec{F}_{AD}$$

$$= (12.31 + 0.29 F_{AC} - 0.28 F_{AD}) \hat{i} + (-36.92 - 0.86 F_{AC} - 0.96 F_{AD}) \hat{j} + (9.23 - 0.43 F_{AC}) \hat{k}$$

For resultant to be vertical, the x and z components must be zero.

$$\text{i.e. } 9.23 - 0.43 F_{AC} = 0$$

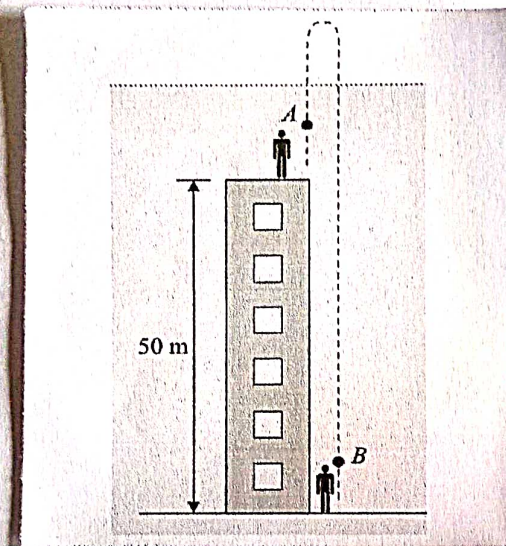
$$F_{AC} = \frac{9.23}{0.43} = \underline{\underline{21.46 \text{ N}}}$$

$$12.31 + 0.29 F_{AC} - 0.28 F_{AD} = 0$$

$$12.31 + 0.29 \times 21.46 - 0.28 F_{AD} = 0$$

$$F_{AD} = \frac{18.53}{0.28} = \underline{\underline{66.19 \text{ N}}}$$

17. A ball is thrown upwards from the top of a 50m high building with an initial velocity of 20m/s. At the same instant, another ball is thrown upwards with an initial velocity of 30m/s from ground. Determine i) when and where they will meet each other, ii) the velocity of each ball at that instant.



Given: -

$$u_A = 20 \text{ m/s}$$

$$u_B = 30 \text{ m/s}$$

$$a = -g = -9.8 \text{ m/s}^2$$

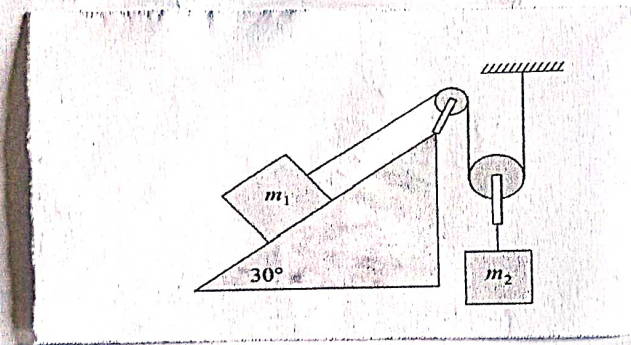
To reach the velocity is zero,

$$v_A = u_A + a_A \cdot t_A$$

$$0 = 20 + (-9.8) \times t_A$$

$$t_A = \underline{\underline{2.03 \text{ s}}}$$

18. Determine the acceleration of system of blocks shown in figure. The coefficient of kinetic friction between block 1 and inclined plane is 0.15. Also determine the tension in string. Take $m_1 = 75\text{ kg}$, $m_2 = 50\text{ kg}$.



Given:-

$$m_1 = 75\text{ kg}$$

$$W_1 = 75 \times 9.81 = 735.75\text{ N}$$

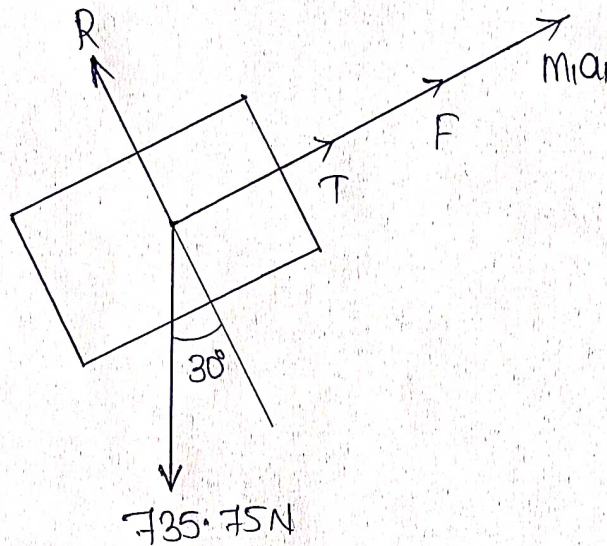
$$m_2 = 50\text{ kg}$$

$$W_2 = 50 \times 9.81 = 490.5\text{ N}$$

$$\mu = 0.15$$

The block m_2 (50kg) will move upwards and block m_1 (75kg) move downwards along the plane.

FBD of m_1 (75kg) block



$$\Sigma F \perp \text{ to the plane} = 0$$

$$R - 735.75 \cos 30 = 0$$

$$R = \underline{\underline{637.17 \text{ N}}}$$

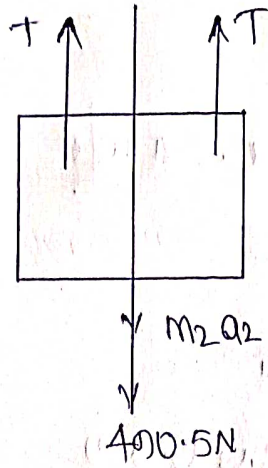
$$\Sigma F \text{ parallel to the plane} = 0$$

$$T + F + 75a_1 - 735.75 \sin 30 = 0$$

$$T = 367.87 - 0.15 \times 637.17 - 75a_1$$

$$T = 272.29 - 75a_1 \quad \text{--- (1)}$$

FBD of $m_2(50\text{kg})$ block



$$\underline{\sum F_y = 0}$$

$$T + T - 490.5 - 50 \times a_2 = 0$$

$$2T = 490.5 + 50a_2$$

$$T = 245.25 + 25a_2$$

$$\text{We know, } a_2 = \frac{a_1}{2}$$

$$\therefore T = 245.25 + 25 \times \frac{a_1}{2}$$

$$T = 245.25 + 12.5a_1 \quad \text{————— (2)}$$

Comparing equations (1) and (2)

$$272.29 - 75a_1 = 245.25 + 12.5a_1$$

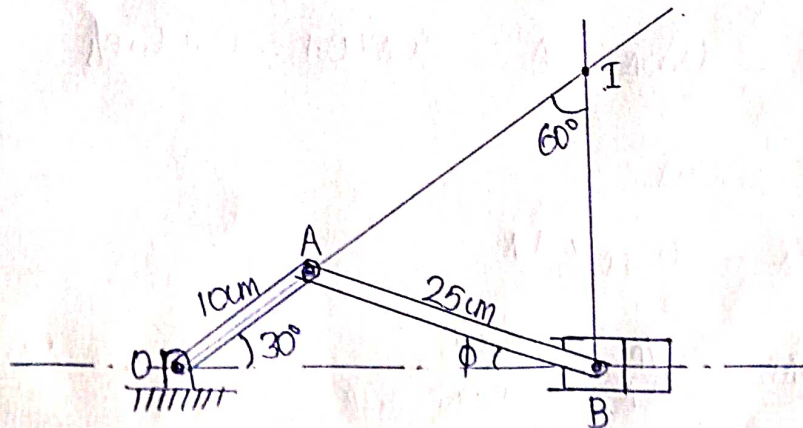
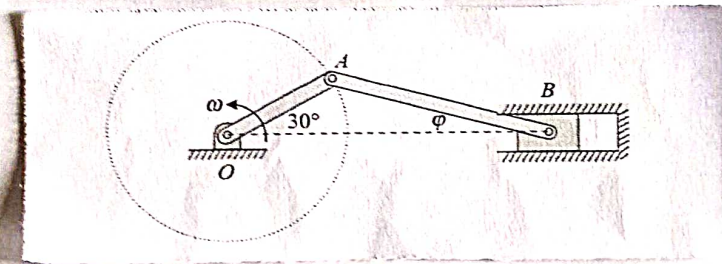
$$27.04 = \cancel{62.5a_1} \quad 87.5a_1$$

$$a_1 = \frac{27.04}{\begin{array}{r} 62.5 \\ 87.5 \end{array}} = \underline{\underline{0.309 \text{ m/s}^2}}$$

$$a_2 = \frac{a_1}{2} = \frac{0.309}{2} = \underline{\underline{0.154 \text{ m/s}^2}}$$

$$\begin{aligned} T &= 272.29 - 75 \times a_1 \\ &= 272.29 - 75 \times \cancel{30} 0.309 \\ &= 272.29 - 23.17 \\ &= \underline{\underline{249.12 \text{ N}}} \end{aligned}$$

10. A reciprocating engine mechanism is shown in figure, in which the crank OA rotates at a constant angular velocity of 1200 rpm in the anticlockwise direction. For the position shown, determine i) angular velocity of connecting rod AB, and ii) the velocity of piston in the engine. Take $OA = 10\text{ cm}$, $AB = 25\text{ cm}$.



Consider ΔOAB ,

$$\frac{OA}{\sin \phi} = \frac{AB}{\sin 30}$$

$$\Rightarrow \frac{10}{\sin \phi} = \frac{25}{\sin 30}$$

$$\sin \phi = \frac{10 \sin 30}{25}$$

$$\phi = 11.54^\circ$$

Now consider ΔABF

$$\frac{AB}{\sin 60} = \frac{IB}{\sin 41.54} = \frac{IA}{\sin 78.46}$$

$$IB = 19.14 \text{ cm}$$

$$IA = 28.28 \text{ cm}$$

$$\omega_{OA} = 1200 \times \frac{2\pi}{60} = 40\pi \text{ rad/s}$$

$$V_A = \omega_{OA} \times OA = 40\pi \times 0.1 = 12.56 \text{ m/s}$$

We know

$$V_A = \omega_{AB} \times IA$$

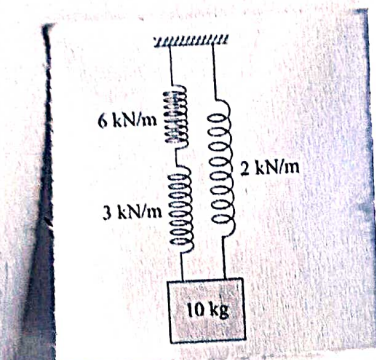
$$\omega_{AB} = \frac{V_A}{IA} = \frac{12.56}{0.2828} = \frac{12.56}{0.2828} = \underline{\underline{44.41 \text{ rad/s}}}$$

$$V_B = \omega_{AB} \times IB = 44.41 \times 0.1914 = \underline{\underline{8.5 \text{ m/s}}}$$

20.

a.

A 10 kg block is suspended by a system of three springs as shown. The respective spring constants are also shown in the figure. Determine the period of oscillation for small displacement from equilibrium position.



Consider the springs in series connection (3 kN/m & 6 kN/m)

$$\frac{1}{k_{e1}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\frac{1}{k_{e1}} = \frac{1}{3} + \frac{1}{6} = \frac{6+3}{18} = \frac{9}{18}$$

$$k_{e1} = 18/9 = 2 \text{ kN/m}$$

Now consider parallel connection (2 kN/m & k_{e1})

$$k_{eq} = 2 + 2 = 4 \text{ kN/m}$$

$$k_{eq} = 4000 \text{ N/m}$$

Time period,

$$\begin{aligned} T &= 2\pi \sqrt{\frac{m}{k}} \\ &= 2\pi \sqrt{\frac{10}{4000}} \\ &= 2\pi \times 0.05 \\ T &= \underline{\underline{0.314 \text{ s}}} \end{aligned}$$

- b. A passenger car is travelling at 65 km/hr on a level road. The distance from road to centre of wheel is 30 cm. Determine
- Magnitude of angular velocity of wheels
 - Magnitude of constant angular deceleration of wheels, if car is brought to rest in 150 m.

Given:-

$$u = 65 \text{ km/h}$$

$$= 65 \times \frac{5}{18} = 18.05 \text{ m/s}$$

i) Angular velocity

$$\omega = \frac{v}{r} = \frac{18.05}{30 \times 10^{-2}} = \underline{\underline{60.16 \text{ rad/s}}}$$